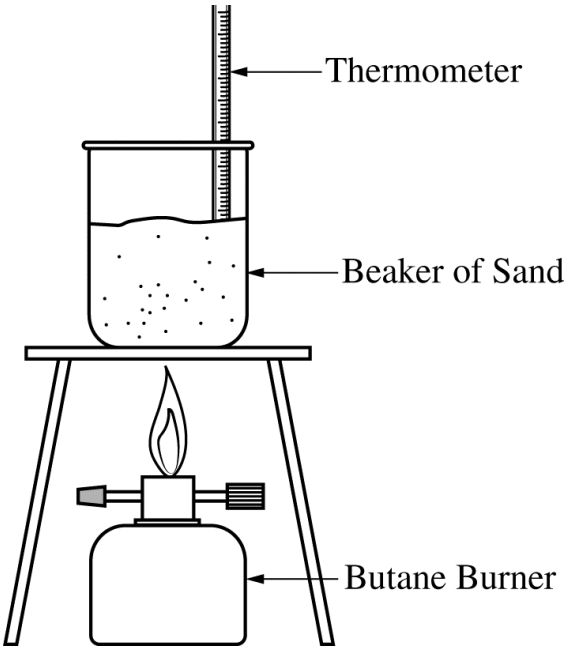
frq1

1. A student is asked to determine what mass of butane, $C_4H_{10}(g)$, needs to burn in order to raise the temperature of a 1650 g beaker of sand by 180.°C. The student is provided with the equipment shown below.



a. Calculate the amount of heat energy needed to raise the temperature of the beaker of sand by 180.°C. Assume that all the heat energy from the burner is transferred to the beaker of sand and that the specific heat capacity of the beaker and sand together is 0.810 J/(g×°C).

The student runs the experiment and collects the data shown in the table below.

Mass of the beaker and sand	1650 g
Mass of butane burner before combustion	225.26 g
Mass of butane burner after combustion	218.20 g
Initial temperature of the beaker of sand	20.°C
Final temperature of the beaker of sand	200.°C

b. Calculate the number of moles of butane that was used in the experiment. Report your answer to the appropriate number of significant figures.

The combustion of butane is represented by the equation below.

$$C_4H_{10}(g) + \frac{13}{2}O_2(g) \rightarrow 5H_2O(g) + 4CO_2(g) \Delta H_{rm}^{\circ} = -2659 \text{kJ/mol}_{rxn}$$

c. Using the balanced equation for the combustion of butane and $\Delta H_{rxn}°$, determine the amount of heat energy produced by the combustion of butane in the experiment.

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d. The student claims that some of the heat energy produced by the combustion of butane was lost to the air surrounding the system. Do your answers to parts (a) and (c) support the student's claim? Explain.

2. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

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Answer the following questions about atomic structure and bonding.

(a) Write the complete ground-state electron configuration of the Kr atom.

(b) The Rb^+ ion has the same electron configuration as the neutral Kr atom. Which has the larger radius, the Rb^+ ion or the Kr atom? Justify your answer in terms of Coulomb's law.

(c) The composition of a sample of solid Rb containing a high percentage of the radioactive isotope Rb-83 is given in the following table.

Isotope	Isotopic Mass (amu)	Percentage of Sample	Isotope Half-Life
Rb-83	82.92	32.5%	86 days
Rb-85	84.91	50.8%	stable
Rb-87	86.91	16.7%	stable

(i) Using the values in the table, calculate the average atomic mass, in amu, of Rb in the sample.

(ii) Rb-83 undergoes radioactive decay into gaseous Kr-83, which leaves the solid sample. Over time, how will the average atomic mass of the remaining sample compare to the value calculated in part (c)(i) ?

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3. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to the analysis of $CaBr_2$.

(a) A student has a $10.0\ g$ sample of $CaBr_2$. Show the setup of the calculation to determine the number of moles of $CaBr_2$ in the sample. Include units in the setup. (You do not need to do any calculations.)

(b) What number, in addition to the answer to part (a), is needed to determine the number of atoms of Ca in the sample?

(c) A different student is given a $10.0\ g$ sample labeled $CaBr_2$ that may contain an inert (nonreacting) impurity. Identify a quantity from the results of laboratory analysis that the student could use to determine whether the sample was pure.

(d) Explain why $CaCl_2$ is likely to have properties similar to those of $CaBr_2$.

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4. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student is asked to prepare a buffer solution made with equimolar amounts of $CH_3NH_2(aq)$ and $CH_3NH_3Cl(s)$. The student uses $25.00\ \text{mL}$ of $0.100\ M\ CH_3NH_2(aq)$, which contains $0.00250\ \text{mol}$ of CH_3NH_2, to make the buffer.

(a) Calculate the mass of $CH_3NH_3Cl(s)$ that contains $0.00250\ \text{mol}$ of CH_3NH_3Cl.

The student has the following materials and equipment available.

• Distilled water	• Electronic balance	• $50\ \text{mL}$ beaker	• Pipets
• $0.100\ M\ CH_3NH_2(aq)$	• Weighing paper	• $10.0\ \text{mL}$ graduated cylinder	• pH meter
• Solid CH_3NH_3Cl	• $50.00\ \text{mL}$ buret	• Small spatula	

(b) The following table contains a partial procedure for making the buffer solution. Fill in steps 1 and 4 to complete the procedure using only materials and equipment selected from the choices given. (Not all materials listed will be used. Assume that all appropriate safety measures are already in place.)

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Step	Procedure
1	
2	Place the solid in the 50mL beaker.
3	Clean the buret and rinse with distilled water.
4	
5	Use the buret to add 25.00mL of $0.100M\text{CH}_3\text{NH}_2(aq)$ to the beaker.
6	Mix well.
7	Check the pH with the pH meter.

The value of K_b for $\text{CH}_3\text{NH}_2(aq)$ is 4.4×10^{-4}, and the pH of the buffer the student prepared is 10.64.

(c) The student prepares a second buffer solution. The student uses $25.00\ \text{mL}$ of $0.050\ M\ \text{CH}_3\text{NH}_2(aq)$ instead of $25.00\ \text{mL}$ of $0.100\ M\ \text{CH}_3\text{NH}_2(aq)$, and half the mass of $\text{CH}_3\text{NH}_3\text{Cl}(s)$ that was used in the first buffer. Is the pH of the second buffer greater than, less than, or equal to the pH of the first buffer? Justify your answer.

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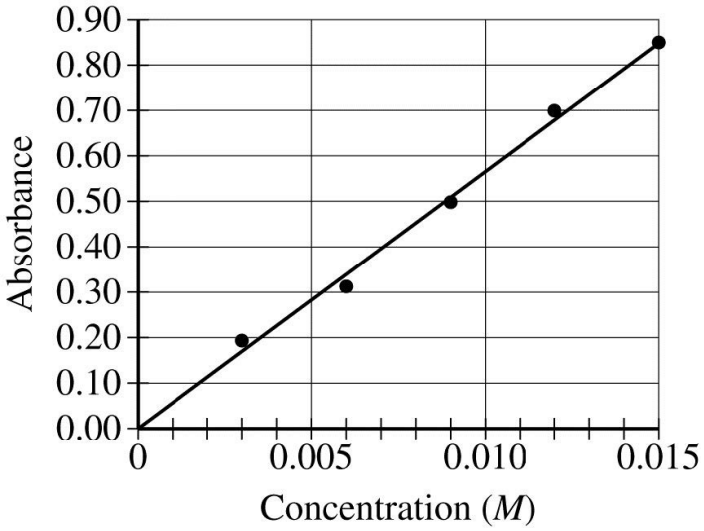
5. To spectrophotometrically determine the mass percent of cobalt in an ore containing cobalt and some inert materials, solutions with known $[Co^{2+}]$ are prepared and the absorbance of each of the solutions is measured at the wavelength of optimum absorbance. The data are used to create a calibration plot, shown below.



A 0.630 g sample of the ore is completely dissolved in concentrated $HNO_3(aq)$. The mixture is diluted with water to a final volume of 50.00 mL. Assume that all the cobalt in the ore sample is converted to $Co^{2+}(aq)$.

a. What is the $[Co^{2+}]$ in the solution if the absorbance of a sample of the solution is 0.74?

b. Calculate the number of moles of $Co^{2+}(aq)$ in the 50.00 mL solution.

c. Calculate the mass percent of Co in the 0.630 g sample of the ore.

frq1

6. **For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

$$4\,\mathrm{Fe}(s) + 3\,\mathrm{O_2}(g) \rightarrow 2\,\mathrm{Fe_2O_3}(s) \qquad \Delta H^\circ = -1650\,\mathrm{kJ/mol}_{rxn}$$

A student investigates a reaction used in hand warmers, represented above. The student mixes $\mathrm{Fe}(s)$ with a catalyst and sand in a small open container. The student measures the temperature of the mixture as the reaction proceeds. The data are given in the following table.

Time (min)	Temperature of Mixture$(^\circ\mathrm{C})$
0	22.0
1	25.1
2	34.6
3	37.3
4	39.7
5	39.4

(a) The mixture ($\mathrm{Fe}(s)$, catalyst, and sand) has a total mass of $15.0\ \mathrm{g}$ and a specific heat capacity of $0.72\ \mathrm{J/(g \cdot {}^\circ C)}$. Calculate the amount of heat absorbed by the mixture from 0 minutes to 4 minutes.

(b) Calculate the mass of $\mathrm{Fe}(s)$, in grams, that reacted to generate the amount of heat calculated in part (a).

(c) In a second experiment, the student uses twice the mass of iron as that calculated in part (b) but the same mass of sand as in the first experiment. Would the maximum temperature reached in the second experiment be greater than, less than, or equal to the maximum temperature in the first experiment? Justify your answer.

frq1

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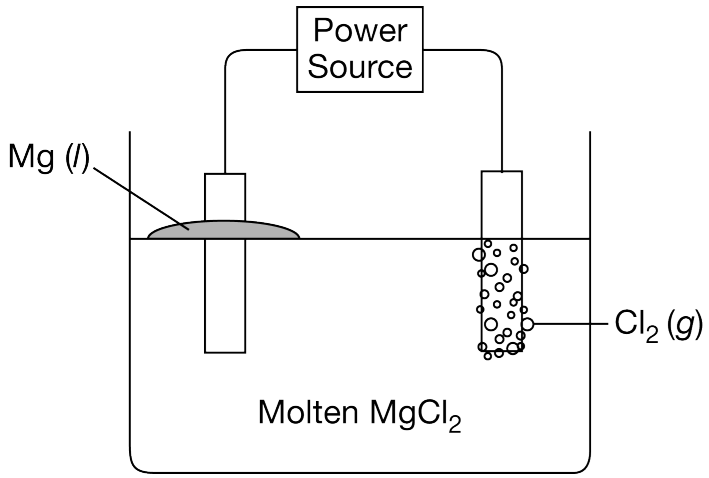

Half-Reaction	$E°$ (V)
$Mg^{2+} + 2\,e^- \rightarrow Mg$	−2.37
$Cl_2 + 2\,e^- \rightarrow 2\,Cl^-$	+ 1.36

Molten $MgCl_2$ can be decomposed into its elements if a sufficient voltage is applied using inert electrodes. The products of the reaction are liquid Mg (at the cathode) and Cl_2 gas (at the anode). A simplified representation of the cell is shown above. The reduction half-reactions related to the overall reaction in the cell are given in the table.

(a) Draw an arrow on the diagram to show the direction of electron flow through the external circuit as the cell operates.

(b) Would an applied voltage of $2.0\ \text{V}$ be sufficient for the reaction to occur? Support your claim with a calculation as part of your answer.

(c) If the current in the cell is kept at a constant 5.00 amps, how many seconds does it take to produce $2.00\ \text{g}$ of $Mg(l)$ at the cathode?

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8. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

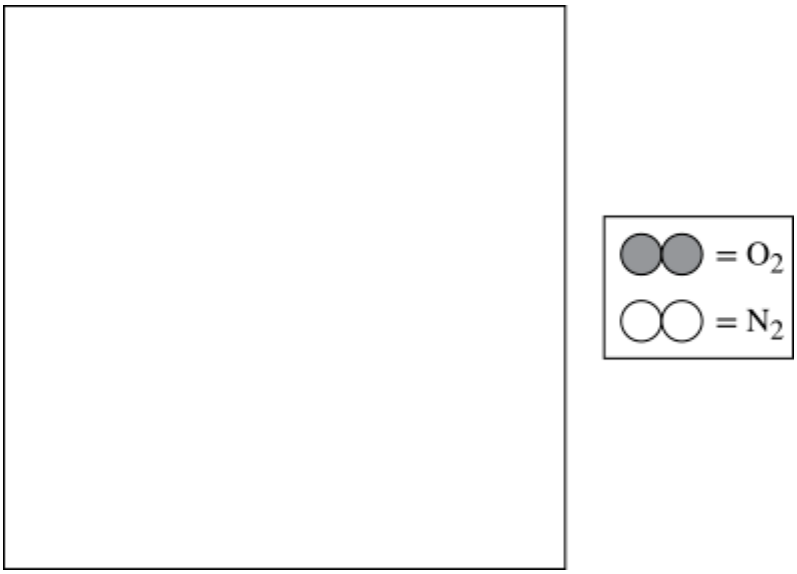
Dry air is comprised of N_2, O_2, and traces of other gases.

A student obtains a $2.00\ L$ container filled with a sample of dry air at a total pressure of $1.30\ atm$ and a temperature of $295\ K$.

(a) Calculate the total number of moles of gas in the container.

(b) The student cools the gaseous mixture until either N_2 or O_2 condenses. Which gas will condense first? Justify your answer in terms of the types and relative strengths of the intermolecular forces in N_2 and O_2.

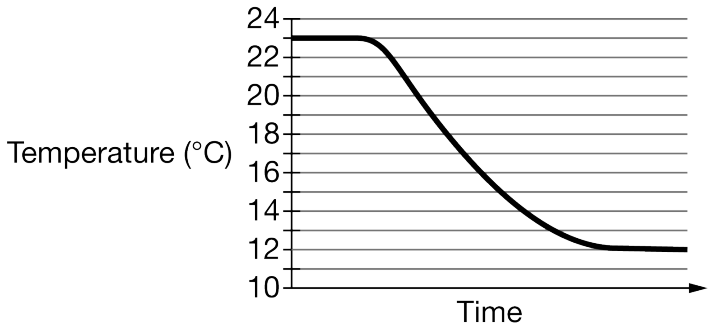
(c) Using the legend given, draw at least three molecules of each type of gas in the sealed container represented below when one of the substances has condensed, but the other substance is still a gas.



(d) In this sample, 78% of the molecules are N_2 and 21% of the molecules are O_2. Calculate the total number of grams of the substance that condenses first.

frq1

9. **For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**



A student conducts an experiment to determine the value of ΔH°_{soln} for the dissolution of $NaC_2H_3O_2(s)$. The student dissolves $10.0\ g$ of $NaC_2H_3O_2(s)$ in room-temperature water in a beaker and measures the temperature over time. The data are given in the graph above.

(a) The student touches the side of the beaker after the dissolution has occurred and observes that it is cold. What experimental evidence is consistent with the student's observation?

(b) Is the dissolution of $NaC_2H_3O_2(s)$ endothermic or exothermic? Justify your answer in terms of the flow of energy between the system and surroundings.

(c) The student calculated the energy change for the dissolution to be $4600\ J$. Based on this value, calculate ΔH°_{soln} in kJ/mol.

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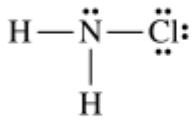
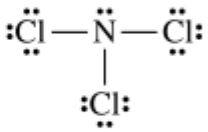
10. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about the compounds NH_2Cl and NCl_3. The Lewis electron-dot diagrams of the two compounds are shown.





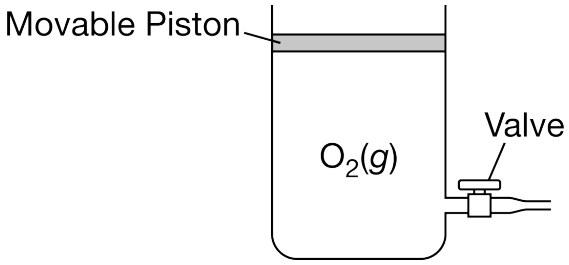
(a) Calculate the number of moles of NH_2Cl (molar mass $51.48\ \text{g/mol}$) present in $1.0\ \text{L}$ of a solution in which the concentration of NH_2Cl is $0.0016\ \text{g/L}$.

(b) NH_2Cl is highly soluble in water, whereas NCl_3 is nearly insoluble. Explain this observation in terms of the types and relative strengths of the intermolecular forces between each of the solutes and water.

(c) The value of $\Delta H^{\circ}_{vaporization}$ for $NCl_3(l)$ is $32.9\ \text{kJ/mol}$. Calculate the amount of energy required to vaporize a $15.0\ \text{g}$ sample of NCl_3 (molar mass $120.36\ \text{g/mol}$).

frq1

11. **For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**



A student investigates gas behavior using a rigid cylinder with a movable piston of negligible mass, as shown in the diagram above. The cylinder contains $0.325\ \text{mol}$ of $O_2(g)$.

(a) The cylinder has a volume of $7.95\ \text{L}$ at 25°C and $1.00\ \text{atm}$. Calculate the density of the $O_2(g)$, in g/L, under these conditions.

(b) Attempting to change the density of the $O_2(g)$, the student opens the valve on the side of the cylinder, pushes down on the piston to release some of the gas, and closes the valve again. The temperature of the gas remains constant at 25°C. Will this action change the density of the gas remaining in the cylinder? Justify your answer.

(c) The student tries to change the density of the $O_2(g)$ by cooling the cylinder to -55°C, which causes the volume of the gas to decrease. Using principles of kinetic molecular theory, explain why the volume of the $O_2(g)$ decreases when the temperature decreases to -55°C.

(d) The student further cools the cylinder to -180°C and observes that the measured volume of the $O_2(g)$ is substantially smaller than the volume that is calculated using the ideal gas law. Assume all equipment is functioning properly. Explain why the measured volume of the $O_2(g)$ is smaller than the calculated volume. (The boiling point of $O_2(l)$ is -183°C.)

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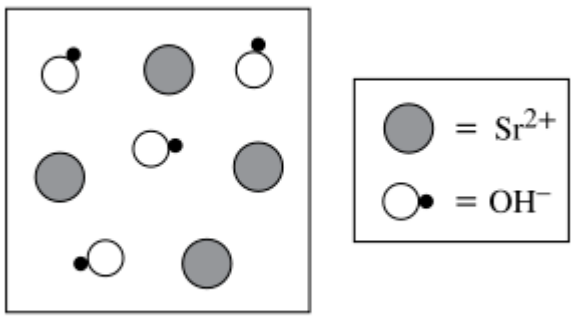
12. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Strontium hydroxide dissolves in water according to the following equation. The K_{sp} expression for strontium hydroxide is provided.

$$\mathrm{Sr(OH)_2}(s) \rightleftarrows \mathrm{Sr^{2+}}(aq) + 2\ \mathrm{OH^-}(aq) \qquad K_{sp} = \left[\mathrm{Sr^{2+}}\right]\left[\mathrm{OH^-}\right]^2$$



(a) A student draws the particulate diagram shown to represent the ions present in an aqueous solution of $\mathrm{Sr(OH)_2}$. (Water molecules are intentionally omitted.) Identify the error in the student's drawing.

(b) The student prepares a saturated solution by adding excess $\mathrm{Sr(OH)_2}(s)$ to distilled water and stirring until no more solid dissolves. The student then determines that $\left[\mathrm{Sr^{2+}}\right] = 0.043\ M$ in the solution.

(i) Calculate the value of $\left[\mathrm{OH^-}\right]$ in the solution.

(ii) Calculate the value of K_{sp} for $\mathrm{Sr(OH)_2}$.

(c) The student prepares a second saturated solution of $\mathrm{Sr(OH)_2}$ in aqueous $0.10\ M\ \mathrm{Sr(NO_3)_2}$ instead of water. Will the value of $\left[\mathrm{OH^-}\right]$ in the second solution be greater than, less than, or equal to the value in the first solution? Justify your answer. (Assume constant temperature.)